

Akshar Patel

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Active DoD Secret Clearance

EDUCATION

Rutgers University, New Brunswick, NJ

Expected Graduation: May 2028

B.S. in Aerospace Engineering (ABET) | GPA: 3.68

- Relevant Coursework: Aerospace Materials, Dynamics, Dynamic Systems and Controls, Fluid Mechanics, Statics, Thermodynamics

TECHNICAL SKILLS

Program Management: Jira, Confluence, Program Increment (PI) Planning, Agile, Budget and Schedule, Risk Management, Gate Reviews

Systems Engineering: Cameo Systems Modeler, SysML, MBSE, Requirements Development, Verification and Validation, Trade Studies

CAD: PTC Creo, Siemens NX, SolidWorks, Fusion 360, Onshape, AutoCAD

Analysis and Simulation: Ansys Mechanical, Ansys Fluent/CFD, Finite Element Analysis (FEA), Creo Simulate, Thermal Desktop

Programming and Tools: Python, MATLAB/Simulink, Java, GitHub, Microsoft Excel, PowerPoint, Word, Slack

EXPERIENCE

Lockheed Martin

Moorestown, NJ

Program Management Apprentice

June 2026 - Present

- Led weekly program status updates to the customer, reporting technical progress, schedule risk, and open actions to align stakeholders
- Directed tasking for 8 subteam engineering POCs on an early start program, tracking deliverables against milestones to hold schedule
- Cut time on manual admin tasks by leading a 3-person team to build AI agents for charging info, onboarding, and cost account tracking
- Co-authored the budget, schedule, and risk assessment package that a business area president evaluated across 5 formal gate reviews
- Lowered Jira ticket entry time 96% by automating issue creation for Program Increment Planning, saving the team hours of manual entry

Mechanical Engineering Apprentice

July 2026 - Present

- Reduced peak circuit board temperature 10°C by simulating convection across 3 fans in Creo Simulate, driving the final fan placement
- Led the initial thermal design by building a 1D thermal circuit model in Excel that mapped heat gradients across 10 board components
- Released 20+ multi-body part drawings to manufacturing by validating each one in PTC Creo for full compliance with drafting standards

Systems Engineering Apprentice

June 2025 - June 2026

- Reduced integration risk across hardware and software interface boundaries by authoring 30 verifiable system requirements in Cameo
- Translated stakeholder needs into a verified solution architecture by modeling system behavior across 10 SysML diagrams in Cameo MBSE

Drifty

Remote | January 2026 - Present

Co-Founder and CTO

- Co-founded a sleep performance startup and own its technical roadmap, leading software development on existing consumer wearables
- Secured \$300K in AWS credits by earning admission to the LvlUp Ventures Labs Accelerator, funding cloud infrastructure for the product
- Grew an audience to 1,000+ followers and 5,000+ views with \$0 in ad spend by owning go-to-market strategy and every campaign launch

Space Ocean Corp.

Remote | January 2026 - June 2026

Lead Thermal Engineer

- Owned the thermal control subsystem for a 12U LEO CubeSat flight demo, from thermal modeling through heater selection
- Simulated transient temperature response over a full LEO orbit by developing a modular 6-node spacecraft thermal model in Python
- Prevented water tank freezing on orbit by validating the heater system through hot and cold case analysis against survival temperatures
- Drove the final heater design by leading 5 thermal trade studies that scored options on 3 criteria: mass, power, and recurring unit cost

PROJECTS

UAS Avionics Enclosure Structural and Thermal Qualification

Fusion 360, FEA, Python

- Achieved a +0.16 margin of safety across an 87 to 2,293 Hz vibration band by running multiple modal FEA frequency sweeps in Fusion
- Proved a sealed 10.8 W avionics enclosure needed 9.2x more cooling area by building a Python heat transfer analysis of the sealed design
- Verified the sized aluminum enclosure by tracing all 15 structural and thermal requirements to component datasheet limits in Fusion 360

Dragon Capsule Aerothermal Re-entry Analysis

Ansys Fluent, Python

- Matched Rayleigh Pitot and modified Newtonian theory within 3% from Mach 5 to 10 by building and validating a reentry model in Fluent
- Stabilized hypersonic solver runs by iterating on the Courant number and ramping each Mach case from a converged lower speed solution
- Found Avcoat 43% heavier among 3 heat shield materials by coupling Fay-Riddell heating to a 1D transient conduction solver in Python